DeepLOB Reproduction and Cross-Market Transfer for Limit Order Book Forecasting

IEOR 242 Spring 2026 Final Project Report

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May 2026

This repository version lists the submitting author only. Full team information and the presentation link can be added in the submission copy.

Abstract

This project studies whether a deep limit order book model can be reproduced on a standard benchmark and then transferred to a materially different market setting. We reproduce DeepLOB on the FI-2010 benchmark [4, 2] using a modern PyTorch pipeline and adapt the same design to Optiver auction-book data [3], where only two book levels are visible. On FI-2010, the strongest short-horizon result reaches accuracy 0.8347 and MCC 0.6080 at 10 events, while the strongest long-horizon result reaches MCC 0.6943 at 100 events under an adaptive training profile. We also compare the deep model against shallow manual-factor baselines and analyze 104 engineered microstructure factors, finding that price and volume derivative features dominate the qualified factor set. For Optiver, we focus on the rolling-quantile label mode and compare zero-shot transfer, per-stock fine-tuning, and specific-stock out-of-sample training. Mean accuracy remains modest, but accuracy is not the correct standalone criterion: per-stock fine-tuning improves Cohen's κ from 0.0618 to 0.0844 and MCC from 0.0626 to 0.0847, even when accuracy declines slightly. Cumulative normalized profit trajectories and a local LIME-style explanation provide complementary evidence about model behavior. The main limitation is persistent nonstationarity: specific-stock out-of-sample training collapses to near-zero agreement, showing that transfer is constrained less by optimization than by distribution shift and label instability.

1 Project Description and Motivation

Limit order book (LOB) forecasting asks whether recent order-flow states contain enough information to predict short-horizon mid-price direction. We study a three-class version of this task: given a recent sequence of LOB states, predict whether the future mid-price will move down, remain stationary, or move up over a fixed number of events. This problem is relevant to execution timing, market making, and short-horizon risk monitoring, but in this project we treat predictions as research diagnostics rather than as a deployable trading signal.

The problem is a good fit for a deep learning project because the input has both spatial and temporal structure. At each event, the bid and ask prices and sizes across levels describe the local state of supply and demand; across time, the evolution of those levels reflects order-flow dynamics. DeepLOB [4] is attractive precisely because it encodes both forms of structure: convolutional layers extract local cross-level patterns, an Inception-style block captures multiple local scales, and an LSTM aggregates short-term temporal context.

The main question is not only whether DeepLOB can be reproduced on FI-2010, but whether its learned inductive bias transfers to a different market representation. That is why the report is organized as a two-stage study: first, benchmark reproduction on FI-2010; second, transfer to Optiver's shallower two-level auction book. This makes the project more than a replication exercise. It becomes an investigation of when deep microstructure representations generalize, when they fail, and what role is played by training control, label design, and fine-tuning. The structure also mirrors the course project requirements directly: a clearly motivated use-case, a precise learning problem, a justified methodology with advanced components, diagnostic discussion of results, and a safety/ethics assessment.

2 Data and Learning Problem

2.1 Dataset Overview

We work with two datasets that share the same prediction target but differ substantially in representation.

FI-2010 benchmark. FI-2010 [2] is a standard LOB benchmark with 40 normalized input features per event, representing 10 bid and ask price levels and their corresponding volumes. We use the standard three-class mid-price movement task at horizons of 10, 20, 30, 50, and 100 events ahead.

Optiver auction-book data. The Optiver workflow is built from processed per-stock files derived from the Optiver realized-volatility competition data [3]. Each event exposes only two order-book levels, yielding 8 reordered features:

$$\left\{ \begin{array}{l} \texttt{ask\_price1}, \texttt{ask\_size1}, \texttt{bid\_price1}, \texttt{bid\_size1}, \\ \texttt{ask\_price2}, \texttt{ask\_size2}, \texttt{bid\_price2}, \texttt{bid\_size2} \end{array} \right\}.$$

All reported Optiver experiments use the 5-event horizon and evaluate transfer on held-out stocks 42, 17, and 82.

Sample construction. Both datasets are converted into overlapping windows. Let $X_t \in \mathbb{R}^{100\times d}$ denote the 100-event lookback window ending at event t, where $d = 40$ for FI-2010 and $d = 8$ for Optiver. The target label is $Y_t \in \{-1, 0, +1\}$, indicating down, stationary, or up movement over a future horizon of k events. The model learns a mapping

$$f_\theta : X_t \mapsto Y_t.$$

2.2 Label Construction and Preprocessing

For FI-2010, we follow the standard benchmark task. For Optiver, labels are constructed from future log mid-price returns using rolling-quantile thresholding: the class boundaries are the rolling historical 33rd and 67th percentiles of absolute k-step returns over a 20,000-sample window. This allows thresholds to adapt to stock-specific volatility scale and reduces the brittleness of fixed absolute thresholds across stocks.

The most important preprocessing design choice is causality. Optiver inputs use rolling normalization computed only from past observations, so each example is standardized without leakage from the future. All train, validation, and test splits are chronological rather than randomly shuffled. This is essential because random mixing would substantially overstate generalization in a nonstationary time-series setting.

2.3 Data Quality and Modeling Challenges

Three issues drive the difficulty of the project.

- **Class imbalance.** Short-horizon labels are often dominated by the stationary class, so accuracy alone can be misleading.
- **Representation mismatch.** FI-2010 exposes 10 levels, whereas Optiver exposes only two. This creates a genuine cross-representation transfer problem.
- **Temporal nonstationarity.** On Optiver, a model trained on one temporal segment of a stock can fail on a later segment of that same stock, even before cross-stock transfer is considered.

3 Methodology

3.1 Model Architecture

DeepLOB on FI-2010. Our FI-2010 model closely follows DeepLOB [4]. The input tensor has shape $(B, 1, T, 40)$ with $T = 100$. The network contains three convolutional blocks for local price-volume structure, an Inception-style module for multi-scale local pattern extraction, a one-layer LSTM with hidden size 64 for temporal aggregation, and a dropout plus fully connected classification head. ReLU is used in hidden layers, and the output logits are mapped to class probabilities through Softmax.

DeepLOBLite on Optiver. For Optiver, we adapt the input interface to $(B, 1, T, 8)$ and change the third convolutional block to use a $(1, 2)$ kernel so the network still aggregates across the narrower spatial axis. The recurrent and classification parts remain similar. This produces a lightweight transfer variant we refer to as DeepLOBLite.

3.2 Objective Function and Training

We train parameters θ by solving

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} L\big(f_\theta(X_i), Y_i\big) + R(\theta).$$

For FI-2010, L is multiclass cross-entropy and optimization uses Adam. The paper-style profile uses learning rate 10^{-3}, batch size 32, and validation-accuracy checkpointing for all horizons. The adaptive profile changes the monitoring rule and regularization by horizon, using stronger regularization and validation-loss monitoring at longer horizons where overfitting appears earlier.

For Optiver, we use focal loss, dropout of 0.30, weight decay of 10^{-4}, gradient clipping at 1.0, and macro-F1-based model selection. The implementation also includes an auxiliary log-return regression head to regularize the shared representation under the smaller two-level input.

3.3 Advanced Components Beyond Class Material

The report includes two methodological elements that go beyond a direct lecture-style baseline.

1. **Rolling-quantile label construction.** Instead of using fixed thresholds, the Optiver task uses rolling historical quantiles of return magnitude. This is a domain-specific response to cross-stock volatility differences.

2. **Horizon-aware training control.** The adaptive FI profile does not use a single monitor everywhere. It changes checkpointing and regularization by horizon to address distinct short-horizon and long-horizon failure modes.

These choices matter because the empirical bottleneck is not simply model capacity. It is calibration under skewed labels, changing return scales, and temporal drift.

3.4 Baselines and Comparisons

On FI-2010, we compare DeepLOB against ridge logistic regression and a two-layer MLP trained on 104 engineered microstructure features, following the spirit of earlier feature-based LOB work [1]. On Optiver, we compare three regimes: universal zero-shot transfer, per-stock fine-tuning, and specific-stock out-of-sample training. These comparisons let us separate the value of the architecture itself from the value of transfer calibration.

4 Experimental Setup

All experiments were run in a cluster-oriented workflow on PSC Bridges-2 using PyTorch. Training was launched through SLURM jobs, while notebooks were used only to summarize saved artifacts.

- **Hardware/runtime.** Long GPU jobs save checkpoints and metrics under `models/` and `results/`.
- **Software.** The workflow uses PyTorch, NumPy, pandas, matplotlib, scikit-learn, SciPy, and statsmodels.
- **Primary metrics.** We report accuracy, Cohen's κ, and MCC. These are all classification metrics, but they emphasize different behaviors.
- **Secondary diagnostics.** We inspect training and validation loss curves, confusion matrices, cumulative normalized profit curves, and a local interpretability plot for a representative Optiver sample.

For this project, the most important evaluation choice is not to over-interpret accuracy. This becomes especially important on Optiver, where modest changes in accuracy can hide meaningful changes in class balance, agreement beyond chance, and directional usefulness.

To keep the main narrative focused, we only visualize representative diagnostic cases in the appendix: the best short-horizon FI run (paper profile, 10 events), the best long-horizon FI run (adaptive profile, 100 events), and the retained Optiver rolling-quantile mode at $k = 5$. Intermediate or less stable cases are summarized numerically in tables rather than given full figure treatment.

5 Results and Discussion

5.1 FI-2010 Reproduction and Benchmark Analysis

Table 1 reports the FI-2010 headline results using the strongest profile for each horizon group.

Table 1: Primary FI-2010 results, using the paper profile for 10–30 events and the adaptive profile for 50–100 events.

Horizon	Selected profile	Accuracy	κ	MCC
10 events	Paper	0.8347	0.5907	0.6080
20 events	Paper	0.7506	0.4879	0.5060
30 events	Paper	0.7726	0.5980	0.6023
50 events	Adaptive	0.7940	0.6748	0.6751
100 events	Adaptive	0.7944	0.6920	0.6943

The most important FI-2010 result is not just that the model performs well, but that *different training rules are best at different horizons*. At short horizons, the paper-style profile is stronger because validation-accuracy checkpointing prevents the model from collapsing toward the stationary class. At long horizons, the adaptive profile is slightly but consistently better because stronger regularization reduces late-stage overfitting. Rather than reproducing all five horizons graphically, we focus on the two clearest cases: the best short-horizon run at 10 events and the best long-horizon run at 100 events. The loss curves in Appendix Figures 1 and 2 support that interpretation: the 10-event run stabilizes after an initially noisy phase, while the 100-event adaptive run stops earlier and keeps a flatter validation profile.

The confusion matrices in Appendix Figures 3 and 4 show why these selected horizons are the right diagnostic examples. At the 10-event horizon the model still faces substantial ambiguity between directional classes and the stationary class, yet it retains meaningful recall on both up and down movements. By 100 events, class separation is visibly cleaner, which is consistent with the rise in both κ and MCC. The intermediate horizons fit the same trend and are therefore summarized in Table 1 rather than discussed at equal length.

The baseline and factor analyses make the contribution more credible. Out of 104 engineered features, 21 survive the qualification pipeline and 20 of them belong to the price-and-volume derivative family. Ridge logistic regression reaches only accuracy 0.710 with near-zero κ, while a two-layer MLP improves to MCC 0.191. DeepLOB reaches accuracy 0.8347 and MCC 0.6080 at the same 10-event horizon, showing that the full temporal architecture extracts structure beyond what a static feature snapshot can recover.

5.2 Optiver Transfer Results

For the Optiver task, we report only the retained rolling-quantile mode in the main text because it produced the clearest and most stable transfer behavior. Table 2 summarizes the results across held-out stocks 42, 17, and 82.

Table 2: Optiver 5-event horizon results under rolling-quantile labels, averaged across held-out stocks 42, 17, and 82.

Regime	Mean accuracy	Mean κ	Mean MCC
Universal zero-shot	0.4211	0.0618	0.0626
Per-stock fine-tuned	0.4110	**0.0844**	**0.0847**
Specific-stock OOS	0.3700	0.0021	0.0021

Table 3: Per-stock Optiver results at the 5-event horizon under rolling-quantile labels.

Stock	Zero-shot		Fine-tuned		Specific OOS
	Acc.	κ	Acc.	κ	κ
42	0.4214	0.0697	0.4044	0.0752	-0.0048
17	0.4181	0.0388	0.4033	0.0841	0.0066
82	0.4239	0.0769	0.4252	0.0938	0.0045

The first rigorous conclusion is that **accuracy is not the right standalone summary for Optiver**. Fine-tuning slightly decreases mean accuracy from 0.4211 to 0.4110, yet improves both κ and MCC materially. That means the fine-tuned model is matching the class distribution more faithfully and achieving better agreement beyond chance, even though it is no longer optimizing the most frequent class as aggressively. In other words, a small drop in accuracy can still correspond to a better classifier under skewed or drifting label distributions. This is exactly the kind of case where a holistic evaluation is more informative than a single headline number.

The base loss and confusion matrix in Appendix Figures 5 and 6 support this point. Optimization is stable, so the problem is not simple failure to fit. The confusion matrix shows that the universal model predicts all three classes, but still over-predicts the stationary label. Per-stock fine-tuning mainly acts as a calibration step that reduces this mismatch. Appendix Figure 7 shows the same pattern in aggregated form: fine-tuning improves agreement over zero-shot transfer, while specific-stock out-of-sample training collapses, indicating that within-stock temporal drift is a more severe challenge than cross-stock adaptation alone. We therefore keep the visual focus on the transfer mode and avoid spending comparable space on the less stable specific-stock regime.

5.3 Why Accuracy Alone Is Insufficient on Optiver

Accuracy is especially fragile here for three reasons. First, it depends strongly on the instantaneous class mix, which changes across stocks and time. Second, the rolling-quantile labels are designed to stabilize return-scale differences rather than to maximize raw accuracy. Third, a conservative classifier can preserve accuracy by leaning toward the stationary class even when it gives poor directional agreement. For those reasons, we treat κ and MCC as primary evaluation metrics on Optiver and treat accuracy as supplementary.

The cumulative normalized profit plot in Appendix Figure 8 provides a third type of evidence. It shows six trajectories: zero-shot and transfer performance on each of the three held-out stocks. The curves are positive for both regimes on all three stocks, but the relative ranking is mixed. Transfer finishes above zero-shot on stocks 42 and 82, while stock 17 shows a lower ending curve despite a stronger κ after fine-tuning. This is exactly why no single scalar metric should dominate the interpretation. The profit proxy is not a tradable PnL claim and does not include transaction costs, but it is still useful as a directional diagnostic on held-out windows. Taken together with κ and MCC, it supports the conclusion that transfer improves calibration and directional consistency without guaranteeing uniform gains under every downstream proxy.

5.4 Optiver LIME-Style Local Explanation

The current version of the report adds a local explanation for the Optiver model using the saved LIME-style interpretability output. Appendix Figure 9 focuses on a representative stationary prediction from stock 82 at horizon $k = 5$, showing the input window together with local importance heatmaps for the base model and the fine-tuned model. Because the saved artifact is a local-importance heatmap rather than the most classical sparse LIME bar chart, we interpret it as a qualitative explanation rather than a definitive causal attribution.

Two points stand out. First, the explanation is concentrated on a subset of top-of-book channels rather than being uniformly spread across the full lookback window, which is consistent with the idea that only certain local price and size transitions are informative for short-horizon classification. Second, the post-transfer explanation is smoother and less dominated by a single block of the input than the base-model explanation. This supports the interpretation that fine-tuning is acting primarily as calibration and reweighting of the existing representation, not as a wholesale relearning of the signal structure. The LIME analysis therefore complements the metric tables: the transfer gain appears to come from better alignment to stock-specific local patterns, not from simply fitting harder.

6 Safety, Security, and Ethics

If a model like this were deployed at scale, the key risk would not be benchmark accuracy but failure under regime shift. A model that appears predictive in one period may become unstable in another, especially in markets with changing liquidity, event intensity, or volatility. In this project, the near-zero specific-stock out-of-sample results are a direct warning sign: the model can appear calibrated on one segment and fail badly on the next.

Security concerns are also concrete. Financial data pipelines are vulnerable to preprocessing mistakes, temporal leakage, and silent distribution drift. Because our Optiver workflow relies on causal normalization and rolling-quantile labels, even a small violation of temporal ordering could make the results look much stronger than they truly are. Reproducibility, audit trails, and strict chronological validation are therefore part of the security story, not just the engineering story.

There are also fairness and misuse concerns. Access to low-latency data and large-scale compute is uneven, so advanced market models can reinforce structural inequalities in who benefits from automation. If a model like this were presented as decision support without uncertainty communication, less sophisticated users could be exposed to disproportionate risk. For that reason, the present results should be interpreted as a research study of representation learning under shift, not as evidence that the model is ready for live deployment.

Reasonable mitigations would include abstention near the decision boundary, explicit drift monitoring, conservative deployment thresholds, and human oversight. Given the current Optiver transfer results, the correct scientific conclusion is caution: the model is informative, but not deployment-ready.

7 Presentation Summary

The accompanying presentation should emphasize the following flow:

1. motivate short-horizon LOB prediction as a structured deep learning problem;
2. explain the difference between FI-2010 benchmark reproduction and Optiver transfer;
3. present DeepLOB and DeepLOBLite, together with rolling-quantile labels and horizon-aware training control;
4. summarize the FI benchmark results and the Optiver transfer tables, while highlighting why Optiver should not be judged by accuracy alone;

5. close with the main limitation: temporal nonstationarity remains the dominant obstacle to reliable transfer.

8 Conclusion

This project shows that DeepLOB can be reproduced credibly on FI-2010 and partially transferred to a different market representation, but the reasons for success differ across datasets. On FI-2010, the architecture itself is strong and the main lesson is that training control matters: the paper-style profile dominates at short horizons, while the adaptive profile is slightly better at long horizons. On Optiver, the architecture still captures useful signal, but the decisive improvements come from label design and calibration through per-stock fine-tuning.

The broader lesson is methodological. For real financial sequence data, model architecture is only one part of performance. Label construction, evaluation metrics, and temporal validation protocol matter just as much. That is why the main contribution of the project is not a claim of strong trading accuracy, but a clearer understanding of what transfers, what does not, and how to evaluate a deep model responsibly under nonstationarity. Under the course rubric, the value of the project comes from both the technical results and the disciplined way those results are analyzed.

A Appendix: FI-2010 Diagnostic Figures

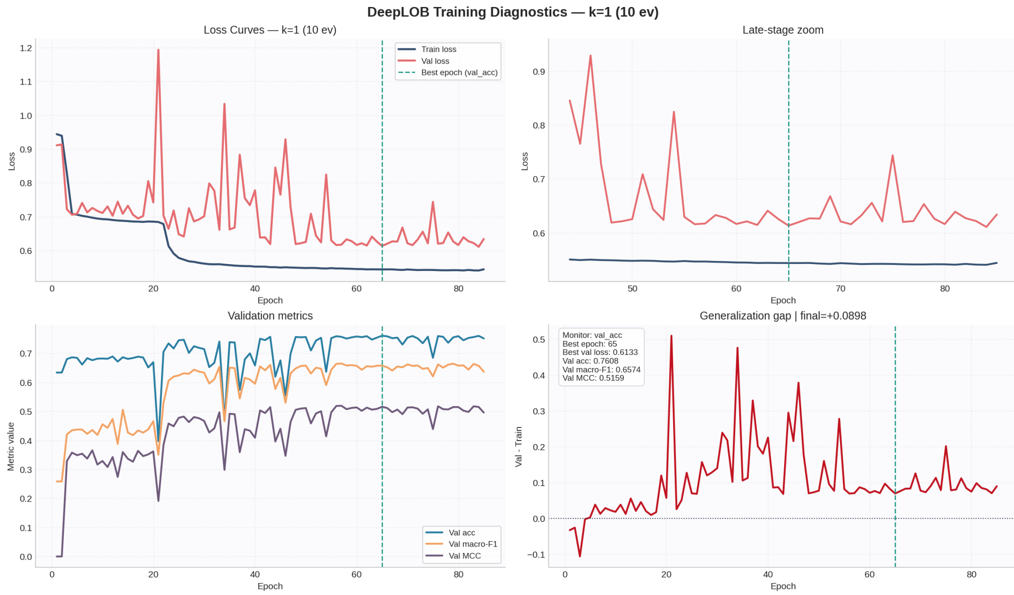

Figure 1: FI-2010 training and validation loss at the 10-event horizon under the paper profile. This is the strongest short-horizon loss curve and supports the claim that the main short-horizon issue is checkpointing and class balance rather than optimization failure.

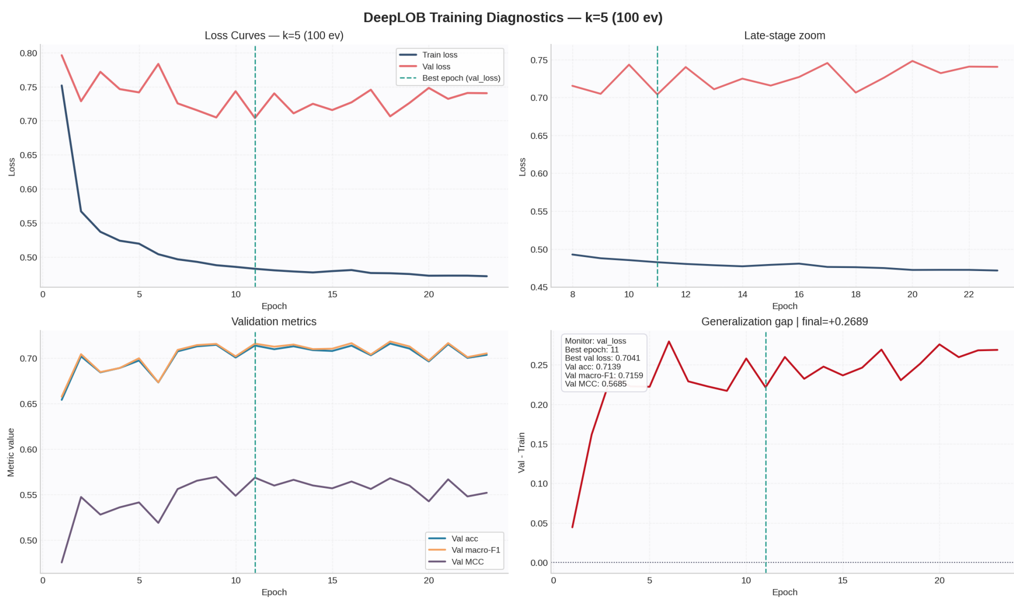

Figure 2: FI-2010 training and validation loss at the 100-event horizon under the adaptive profile. Relative to the paper-style configuration, the adaptive profile reaches a flatter validation trajectory and stronger long-horizon agreement metrics.

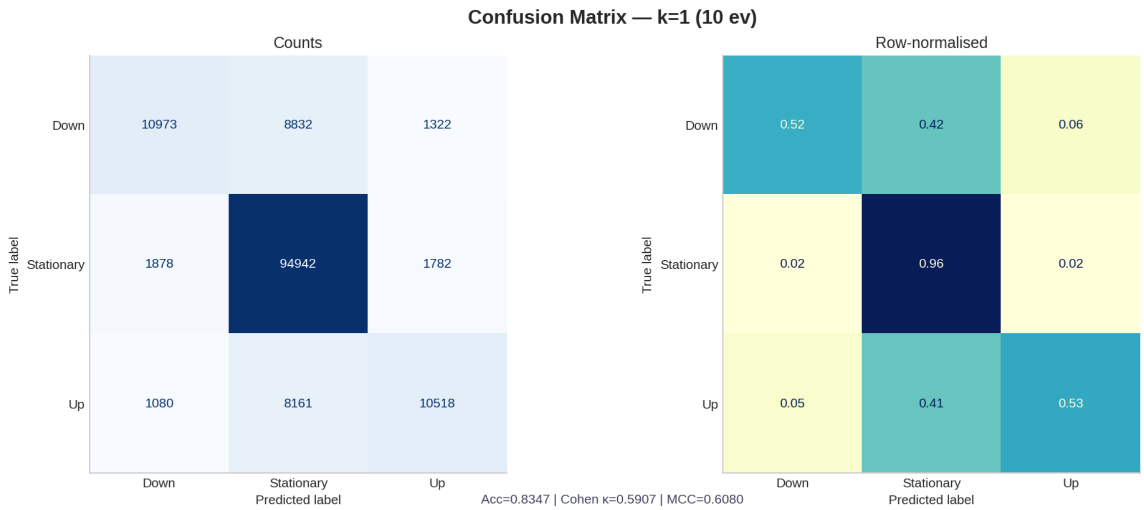

Figure 3: FI-2010 confusion matrix at the 10-event horizon under the paper profile. Even at the hardest horizon, the model preserves meaningful directional recall instead of collapsing entirely to the stationary class.

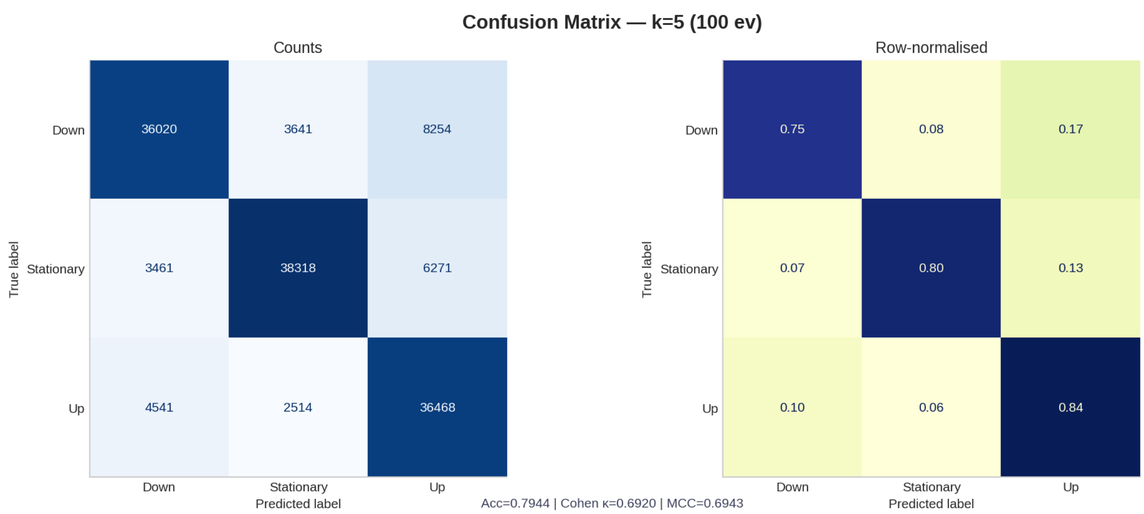

Figure 4: FI-2010 confusion matrix at the 100-event horizon under the adaptive profile. The stronger diagonal structure is consistent with the higher κ and MCC at long horizons.

B Appendix: Optiver Diagnostic Figures

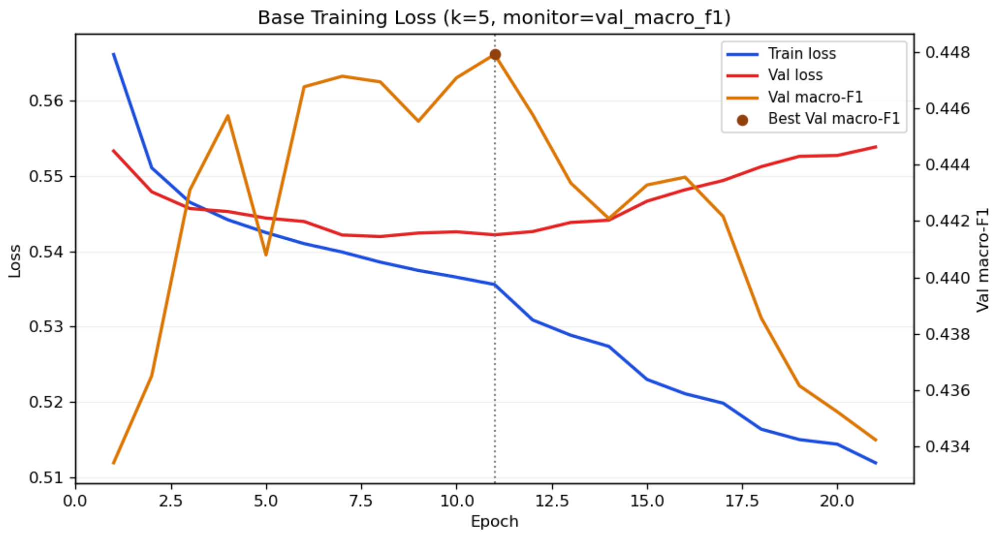

Figure 5: Optiver universal base-model loss curve at the 5-event horizon. Optimization is stable, indicating that transfer limitations are driven more by distribution mismatch than by divergence during training.

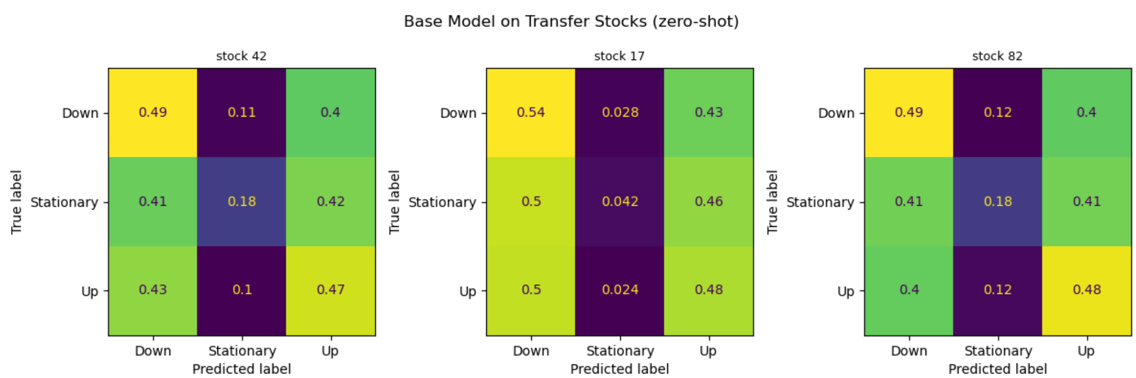

Figure 6: Optiver universal base-model confusion matrix on held-out stocks. The model predicts all classes but still overweights the stationary class, motivating the use of κ and MCC in addition to accuracy.

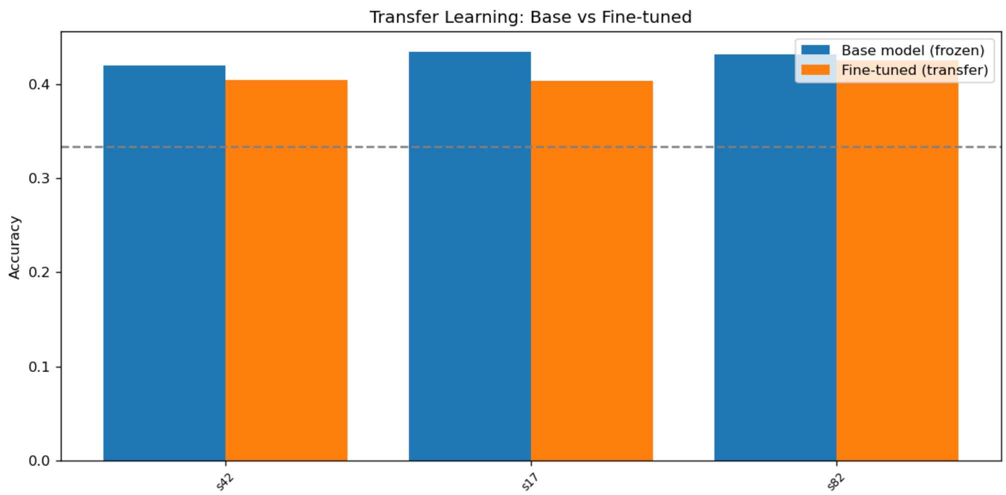

Figure 7: Comparison of Optiver transfer regimes. Fine-tuning improves agreement over zero-shot transfer, while specific-stock out-of-sample training collapses toward zero agreement.

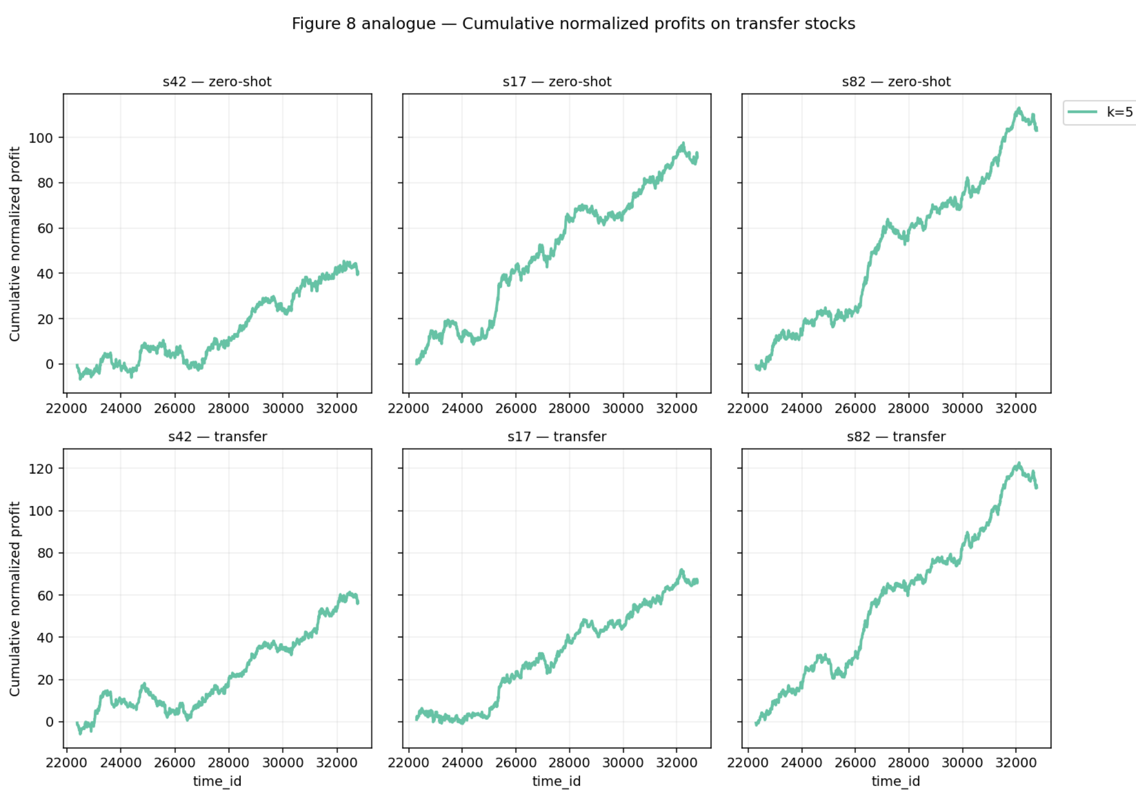

Figure 8: Cumulative normalized profit trajectories for the three held-out stocks under zero-shot transfer and per-stock fine-tuning. These curves are diagnostic only and do not represent executable, transaction-cost-adjusted trading PnL.

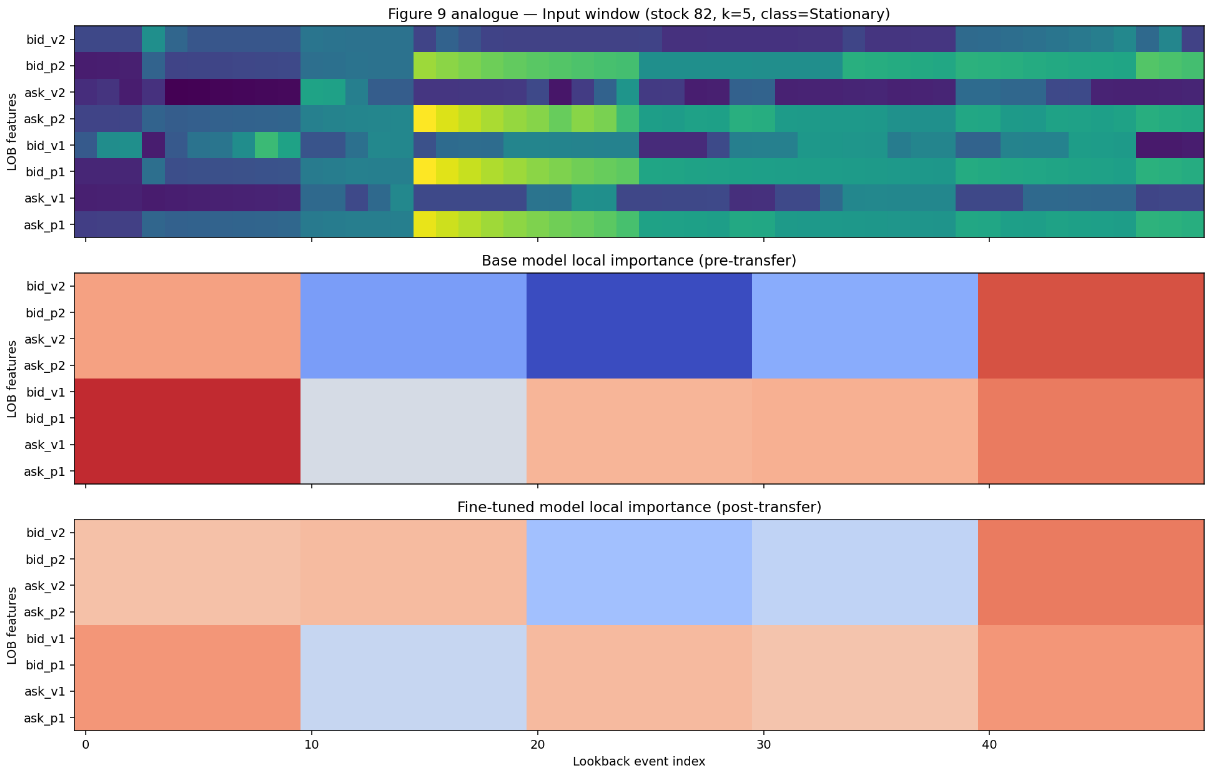

Figure 9: LIME-style local explanation for a representative stationary Optiver sample from stock 82 at horizon $k = 5$. The top panel shows the input window; the lower panels compare the local importance structure before and after transfer fine-tuning.

References

[1] Alec N. Kercheval and Yuan Zhang. Modelling high-frequency limit order book dynamics with support vector machines. *Quantitative Finance*, 15(8):1315–1329, 2015.

[2] Apostolos Ntakaris, Mikko Magris, Juho Kanniainen, Moncef Gabbouj, and Alexandros Iosifidis. Benchmark dataset for mid-price forecasting of limit order book data. *Journal of Forecasting*, 37(8):852–866, 2018.

[3] Optiver and Kaggle. Optiver realized volatility prediction. `https://www.kaggle.com/competitions/optiver-realized-volatility-prediction`, 2021.

[4] Zihao Zhang, Stefan Zohren, and Stephen Roberts. Deeplob: Deep convolutional neural networks for limit order books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.